

Broad Range dsDNA, High Sensitivity dsDNA, Broad Range RNA, and High Sensitivity RNA

All Steps are the same whichever quantification method you are using

1. Determine n number for calculating working reagent for assay: Number of samples + 2 (for standards) + % error (usually .5 or 1)
2. Take appropriate standards for your assay out of the 4 degree fridge and let get to room temp while you set up
3. Create mix for working reagent of Qubit Buffer and Quant-IT Reagent, make sure it is from the box/bag for the type of assay you are running
 - n μ l of Quant-IT Reagent
 - $199 * n$ μ l of Buffer
3. Vortex and spin down working reagent mix (if in a 5mL tube use the larger centrifuge with yellow inserts)
4. Set up thin walled Axygen tubes (in black cardboard boxes on each bench): enough for each of your samples plus 2 more for standards. Label appropriately. They fit nicely in the large black rack on the underside
5. Add 190 μ l of master mix to the 2 tubes for the standards
6. Add 199 μ l of master mix to each of Axygen tubes labeled for your sample
7. Vortex and spin down standards (if large bottles no need to spin down). Add 10 μ l of standard 1 to the standard 1 thin walled tube, add 10 μ l standard 2 to the thin walled standard 2 tube
8. Vortex and spin down your sample before addition. If you have RNA it MUST be kept on ice, DNA should as well. Add 1 μ l of sample to its corresponding thin walled sample tube. Be careful to add the 1 μ l directly to solution in the tube. Never rely on the quick pressure of depressing the plunger to get out all of 1 μ l, small volumes often have stronger attraction to stay inside the pipette tip than to go out into air, but if it has another liquid to go into, it will come out easier
9. Vortex and spin down all tubes (use inserts in minifuge for small tubes)
10. The Qubit is touch-screen. Choose appropriate assay in the Qubit and read standard tubes. DNA standards should have a 10 fold difference between S1 and

S2. It is always best practice to record your standards values. These are read in RFU units

11. If you have only a few samples, wait about a minute before reading. If you have a lot of samples, the time it takes to vortex and read samples should be enough. Read each sample tube one at a time. Our settings are 1 μ l of sample and we want the output to be ng/ μ l
12. Read your samples twice: go through all of the samples once then do through them again. The light that flashes on the sample tube increases the temperature of the liquid, so you do not want to read a sample tube twice quickly
13. All components are safe and can be disposed of in the tip buckets or trash